



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI KRISHNA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

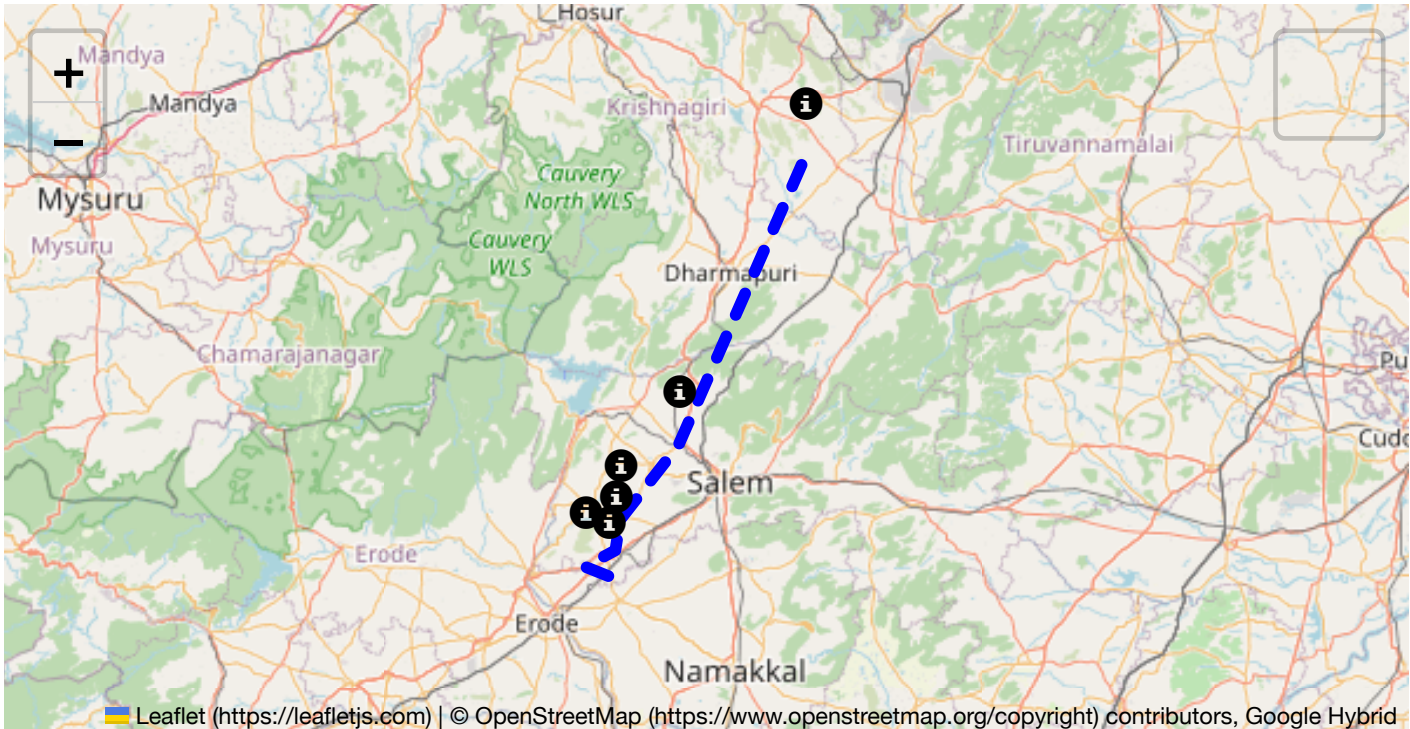
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

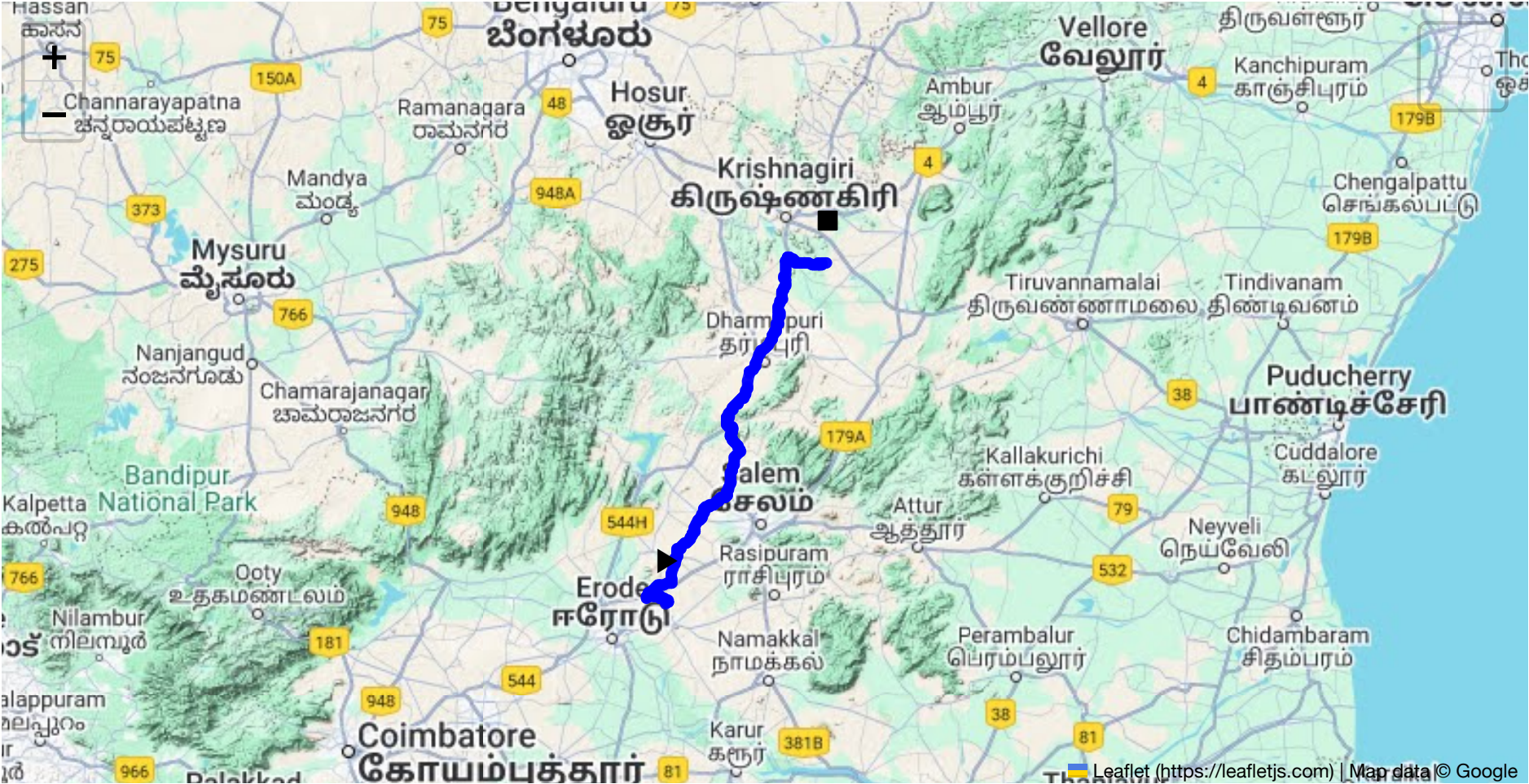
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 152.23 km
Estimated Duration: 3.1 hours
Adjusted Duration (Heavy Vehicle): 3.9 hours

Start: (11.4381, 77.8734)
End: (12.398715, 78.334305)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Karimangalam, covering several waypoints, spans approximately 152.23 kilometers. Starting from Sangagiri, the route follows through Bhavani Main Road, transitioning onto SH 86 and passing through key junctions in areas such as Ayveli, Erumaipatti, and Salem before reaching the destination at Karimangalam-Pochampalli Road. The estimated travel time is around 3.08 hours for trucks carrying hazardous materials.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate with three distinct seasons: a hot summer (March to June), the monsoon season (July to November), and a mild winter (December to February). During the monsoon, heavy rainfall can cause flooding, waterlogging, and reduced visibility. It is crucial for truck drivers to be prepared for slippery roads and potential landslides in hilly areas.

3. Analysis of Traffic Patterns

- Peak Hours:** Early morning (7-9 AM) and late evening (5-8 PM) see significant traffic, particularly near Salem due to its commercial activity.
- Congestion-prone Areas:** Major congestion is expected in Salem city and at key junctions like Bhavani Main Road and SH 86 intersections.

4. Assessment of Road Quality and Infrastructure

Most of the primary roads along this route are well-paved. However, secondary roads may have potholes and uneven surfaces. There are ongoing infrastructural developments, particularly in Salem, where road

widening projects might cause temporary diversions or slow downs.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative to Salem:** For bypassing Salem, consider taking NH 544, although it may extend travel time.
- **Reroute Due to Blockage:** Secondary routes via regional roads in Namakkal district could serve as bypasses but check local road conditions due to potential infrastructure wear.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu has strict regulations on transporting hazardous materials, requiring prior permits. Vehicles must adhere to designated routes and schedules, often avoiding city centers at peak hours.

7. Overview of Historical Incidents

There have been incidents involving chemical spills and fuel tanker overturns in the Salem area. Such areas are typically under scrutiny for rigorous enforcement of safety protocols.

8. Environmental Considerations and Sensitive Areas

- **Protected Areas:** Be cautious when driving near natural reserve borders and agricultural lands to prevent environmental contamination.
- **Urban Sensitivity:** The dense population in Salem requires heightened awareness to prevent accidents.

9. Analysis of Communication Coverage

Mobile network coverage is robust along most of the route, with potential dead zones around remote or rural stretches near Erumaipatti. It is advisable for drivers to have backup communication devices.

10. Estimated Emergency Response Times

- **Sangagiri to Bhavani Main Road:** Approximately 30-40 minutes, owing to moderate road traffic.
- **Salem Vicinity:** Quick response within 20 minutes due to the presence of emergency services.
- **Rural Stretches:** May range from 40 to 60 minutes response time due to accessibility issues.

11. Overall Summary of Risk Assessment

The primary risks on this route involve weather-related hazards, traffic congestion, road quality inconsistencies, and strict regulations pertaining to the transport of hazardous materials. Emergency preparedness, compliance with local laws, route familiarity, and awareness of environmental considerations are essential for minimizing risks.

Truck drivers should remain vigilant, particularly at peak times and in unpredictable weather, and have contingency plans for alternative routing and communication. Regular updates on road conditions and weather forecasts can further aid in safe transport operations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	Medium	11.45546, 77.81800	30 KM/Hr
7	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
8	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
9	Turn	High	11.49469, 77.88488	15 KM/Hr
10	Turn	Medium	11.56072, 77.89919	30 KM/Hr
11	Turn	Medium	11.56140, 77.89991	30 KM/Hr
12	Turn	Medium	11.59085, 77.90592	30 KM/Hr
13	Turn	Medium	11.59253, 77.90690	30 KM/Hr
14	Turn	Medium	11.63848, 77.94825	30 KM/Hr
15	Turn	High	11.64197, 77.95197	15 KM/Hr
16	Turn	Medium	11.64558, 77.95108	30 KM/Hr
17	Turn	Medium	11.67883, 77.96865	30 KM/Hr
18	Turn	High	11.70227, 77.97864	15 KM/Hr
19	Turn	Medium	11.71386, 77.99552	30 KM/Hr
20	Turn	Medium	11.71735, 78.00019	30 KM/Hr
21	Turn	Medium	11.72765, 78.02291	30 KM/Hr
22	Turn	High	11.73785, 78.03496	15 KM/Hr
23	Turn	High	11.73653, 78.03886	15 KM/Hr
24	Turn	Medium	11.74252, 78.04705	30 KM/Hr
25	Turn	Medium	11.77998, 78.05786	30 KM/Hr
26	Turn	Medium	11.86915, 78.08382	30 KM/Hr
27	Turn	Medium	11.87998, 78.08170	30 KM/Hr
28	Turn	Medium	11.91266, 78.05852	30 KM/Hr
29	Turn	Medium	11.92653, 78.06517	30 KM/Hr
30	Turn	Medium	11.93746, 78.05331	30 KM/Hr
31	Turn	Medium	11.95531, 78.04698	30 KM/Hr
32	Turn	High	11.95854, 78.04990	15 KM/Hr
33	Turn	High	11.96015, 78.04945	15 KM/Hr
34	Turn	Medium	11.96266, 78.05115	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
35	Turn	High	11.96433, 78.05113	15 KM/Hr
36	Turn	High	11.96542, 78.04938	15 KM/Hr
37	Turn	High	11.96650, 78.04927	15 KM/Hr
38	Turn	Medium	12.17041, 78.16823	30 KM/Hr
39	Turn	Medium	12.22515, 78.19158	30 KM/Hr
40	Turn	Medium	12.22995, 78.19466	30 KM/Hr
41	Turn	Medium	12.24547, 78.19501	30 KM/Hr
42	Turn	Medium	12.24685, 78.19561	30 KM/Hr
43	Turn	Medium	12.25120, 78.20010	30 KM/Hr
44	Turn	Medium	12.25442, 78.20015	30 KM/Hr
45	Turn	Medium	12.32012, 78.20734	30 KM/Hr
46	Blind Spot	Blind Spot	12.40698, 78.21763	10 KM/Hr
47	Turn	Medium	12.40631, 78.22021	30 KM/Hr
48	Turn	Medium	12.40690, 78.22142	30 KM/Hr
49	Turn	High	12.40667, 78.22287	15 KM/Hr
50	Turn	High	12.41697, 78.22490	15 KM/Hr
51	Turn	High	12.41720, 78.22537	15 KM/Hr
52	Turn	High	12.39684, 78.30837	15 KM/Hr
53	Turn	Medium	12.39491, 78.31228	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
8	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
9	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
18	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
19	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
31	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56072, 77.89919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56140, 77.89991



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59085, 77.90592



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59253, 77.90690



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63848, 77.94825



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64197, 77.95197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64558, 77.95108



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67883, 77.96865



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71386, 77.99552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71735, 78.00019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72765, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73785, 78.03496



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73653, 78.03886



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74252, 78.04705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77998, 78.05786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86915, 78.08382



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87998, 78.08170



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91266, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92653, 78.06517



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



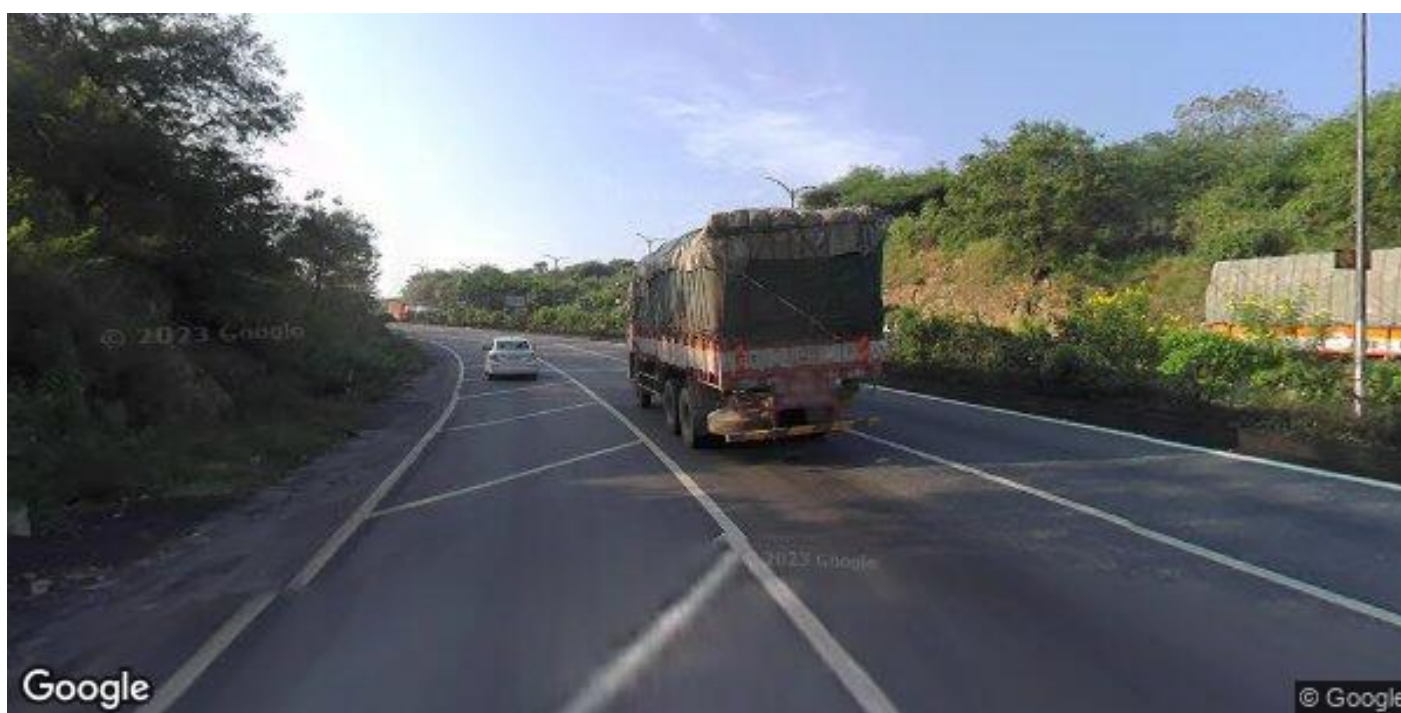
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96266, 78.05115



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.17041, 78.16823



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22515, 78.19158



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22995, 78.19466



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24547, 78.19501



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24685, 78.19561



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25120, 78.20010



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25442, 78.20015



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.32012, 78.20734



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.40698, 78.21763



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40631, 78.22021



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40690, 78.22142



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.40667, 78.22287



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.41697, 78.22490



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.41720, 78.22537

